

Validating and Extending a Chapter-Aligned Moodle Engagement Metric Across STAT0002 and STAT0004

Evidence from online, hybrid and in-person Moodle delivery cohorts

Research Internship Report
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Abstract

Virtual learning environments record how students interact with course materials, offering a scalable basis for monitoring online behavioural engagement. Johnston et al. (2025) developed a chapter-aligned real-time metric and validated it on structured lecture modules; this report applies that fixed measure to five STAT0002 and STAT0004 cohorts at University College London—spanning online, hybrid and in-person periods not used in the original study—and asks whether the association travels, which engagement components matter most, whether engagement affects all students equally, and how practitioners should use the signal safely. Cumulative engagement remains moderately associated with final grade when cohorts are pooled ($r = 0.32$). **Component decomposition** (LMG relative-importance analysis) shows that Diversity and Immediacy account for more unique explained variance than Frequency (pooled shares at week 11: 38%, 36% and 27% respectively), even though all three correlate similarly with grade in bivariate analysis—breadth and promptness of resource use outweigh raw click volume. **Quantile regression** shows that the engagement–grade slope is steeper among lower-performing students than at the median in every cohort (mean excess at $\tau = 0.1$ versus $\tau = 0.5$: +2.3 grade points per standard deviation of engagement), supporting targeted support for at-risk learners rather than a uniform predictor across the ability distribution. Univariate quantile models identify Immediacy as the component with the strongest lower-tail association. Trajectory classes and early-warning precision are reported as supporting diagnostics. Delivery-period comparisons are descriptive because delivery mode is confounded with cohort and academic year. The metric suits low-cost supportive monitoring with local validation, not high-stakes prediction.

1 Introduction

Student engagement—the time and effort students devote to learning activities—is widely linked to academic performance, satisfaction and retention. In online and blended higher education, much of the *behavioural* dimension of engagement can be observed through virtual learning environment (VLE) log data: which resources students access, how often they return, and how promptly they engage with newly released material. Moodle and similar platforms therefore provide an unobtrusive, scalable record of online study behaviour that is increasingly used in learning analytics and early student support (Henrie et al., 2015).

Institutions nonetheless face a practical tension. Threshold-based early-warning tools depend on cut-offs that are difficult to justify in advance, while predictive models often require historical outcome data and may fail to transfer across cohorts or courses. An alternative is to construct interpretable composite indicators grounded in engagement theory and computable in real time from log data alone. Johnston et al. (2025) developed such a measure by reconceptualising the

course-wide engagement score of Chong and Wong (2019) into a weekly, chapter-aligned formulation based on Frequency, Immediacy and Diversity (F, I, D). Their study showed that the metric aligns with the retrospective course-wide score from early in the term, relates to final grades in structured lecture-based modules, and can help identify low-performing students through engagement quintiles—though with the modest precision typical of early-warning systems.

Johnston et al. (2025) establish that a chapter-aligned engagement metric can be computed in real time and can relate to final grades in structured modules. The present report takes that metric as fixed and asks a different question: how informative, robust and interpretable it is when applied to additional STAT0002 and STAT0004 cohorts spanning online, hybrid and in-person delivery periods? The contribution is therefore not metric development, but secondary validation and interpretive extension. Two questions receive particular emphasis. **First**, which aspects of engagement—Frequency, Immediacy and Diversity—are most strongly associated with academic performance once their overlap is accounted for? Relative-importance (LMG) decomposition addresses this under the high collinearity among the three indicators. **Second**, does engagement affect all students equally, or is the association stronger among lower-performing students? Quantile regression examines the engagement–grade relationship across the grade distribution rather than at the average alone. Additional analyses pool evidence across cohorts, test temporal regularity and engagement trajectories, and examine robustness and practical limits on student support.

The report is organised around these priorities. Section 4.2 develops the component-importance question in detail; Section 4.3 develops the quantile-regression question. Broader validation (pooled association, quintile separation), trajectory diagnostics, early-warning precision and robustness checks provide supporting context. Delivery-period comparisons are treated as descriptive boundary checks rather than causal tests.

2 Background and metric context

Online behavioural engagement refers to observable, sustained interaction with learning activities in a digital environment. Engagement theory distinguishes behavioural, emotional and cognitive forms of involvement; Moodle logs capture primarily the behavioural dimension—participation, effort and attention to academic tasks—rather than motivation or understanding (Fredricks et al., 2004; Henrie et al., 2015). It is only a *proxy* for learning: logs record what happens inside the VLE, not offline study, in-person attendance or the quality of students’ reasoning. Even so, when behavioural engagement is measured coherently and validated against academic outcomes, it can support timely monitoring and targeted support.

The course-wide metric of Chong and Wong (2019) combines five indicators—Immediacy, Frequency, Diversity, Recency and Interval—into a single retrospective score aggregated across an entire course. That design is useful for end-of-term analysis but poorly suited to weekly monitoring, and it treats the course as homogeneous rather than aligned with its chapter structure. Johnston et al. (2025) addressed this by computing Frequency, Immediacy and Diversity separately for each chapter at each week, min-max scaling each indicator relative to peers at that point in the term, and summing them into a chapter-level score $IDF = F + I + D$. Chapter scores are combined across released chapters to yield a cumulative engagement measure that can be updated as the term progresses without outcome data or model training. Recency and Interval were excluded because they become unstable when measured mid-course.

Chapter alignment matters because engagement is most interpretable when measured in step with how material is released. For early-warning use, practitioners also need to understand the recall–precision trade-off: flagging the lowest-engagement students may capture many who ultimately struggle, but also many who would have succeeded without intervention (Conijn et al., 2017). The present report takes the metric definition as given and focuses on how it behaves on

new UCL cohorts and what additional diagnostic insight can be extracted from its components and related temporal patterns.

3 Data

The analysis uses five module-year deliveries from two undergraduate statistics modules at University College London. STAT0002 is a first-year foundations module represented here by an online cohort (2020–21) and an in-person cohort (2022–23). STAT0004 is a second-year module and the only module in this dataset spanning all three delivery conditions: online (2020–21), hybrid (2021–22) and in-person (2022–23). Each delivery covers eleven teaching weeks with up to ten labelled chapters. Final grades are recorded on a 0–100 scale; grades below 50 were treated as low performance, and grades below 40 as failure. Students with a recorded grade of zero, typically reflecting exam absence, were excluded.

Table 1 summarises the cohorts. The main engagement–grade sample comprises $N = 1,291$ students with both engagement records and valid final grades after exclusions. The latent trajectory analysis below uses $N = 1,290$ students with complete weekly engagement profiles suitable for class assignment; the difference of one student reflects missing weekly coverage in the trajectory model, not an error in the cohort inventory.

Table 1: Cohort inventory for the main engagement–grade sample ($N = 1,291$).

Module	Year	Condition	N	Mean (SD)	% < 50	% < 40
STAT0002	2020–21	Online	342	66.8 (13.8)	9.9	1.8
STAT0002	2022–23	In-person	268	67.4 (13.0)	10.4	2.6
STAT0004	2020–21	Online	312	62.7 (12.2)	12.5	2.9
STAT0004	2021–22	Hybrid	172	71.9 (8.8)	1.7	0.0
STAT0004	2022–23	In-person	197	70.6 (11.3)	3.6	1.0
Total			1,291			

Source: author’s analysis of UCL Moodle log data.

Two design features constrain interpretation. First, the share of low-performing students varies widely across cohorts (from 1.7% to 12.5%), which limits how sharply any early-warning flag can perform in the highest-achieving deliveries. Second, delivery condition is **perfectly confounded** with academic year and cohort: online corresponds to 2020–21, hybrid to 2021–22, and in-person to 2022–23. Comparisons across conditions therefore describe different module-year contexts rather than randomly assigned delivery modes, and no causal claim about online, hybrid or in-person delivery can be supported.

The modules also differed in assessment structure. STAT0002 in-person records separate exam, participation and peer-marking components; STAT0004 deliveries include coursework and quiz marks, with a group component in the in-person year. This matters because Moodle engagement may align more closely with individual or exam-style assessment than with group, peer-marked or participation components. Where component marks were available, this report therefore analyses whether the engagement metric correlates more strongly with individually assessed work than with collaborative or peer-assessed components.

The two modules also differ in the strength of the engagement–grade link across deliveries. STAT0002 cohorts show end-of-term correlations of $r = 0.28$ – 0.41 , whereas STAT0004 ranges from $r = 0.19$ to $r = 0.30$. As interpretive context, the two modules may differ in ways that affect how well Moodle activity proxies study behaviour; several factors may contribute, though they cannot be disentangled in this observational design. STAT0002 is a first-year foundations module with substantial individually assessed exam weight, where sustained access to lecture

materials and problem sheets may track preparation more directly. STAT0004 is a second-year module with greater student autonomy, more coursework and group-assessment weighting, and possibly more study away from Moodle—each of which would weaken the observable link between VLE logs and final outcomes. How tightly Moodle resources are organised by chapter may also matter: where chapter labelling aligns less clearly with assessed work, a chapter-aligned metric would legitimately carry less outcome signal, as Johnston et al. (2025) report for less structured courses.

4 Engagement feature construction

Weekly chapter-level contribution values were available for each student. These values are stored as *per-week increments*, not as cumulative totals. For each student and week, contributions were summed across chapters to obtain weekly Frequency, Immediacy and Diversity totals. The weekly composite was then defined additively as

$$\text{IDF}_{\text{week}} = \text{freq}_{\text{week}} + \text{imm}_{\text{week}} + \text{div}_{\text{week}},$$

and cumulative measures were obtained by summing across weeks: `cum_freq`, `cum_imm`, `cum_div` and `cum_IDF`. This additive definition matches the chapter-level formulation of Johnston et al. (2025); it is not a multiplicative composite. A reconstruction check confirmed that the supplied contributions are consistent with the intended Frequency and Diversity definitions.

For extension analyses, additional timing and regularity features were derived from event logs (for example, variation in weekly activity and gaps between sessions). Latent-class mixed models were also fitted to normalised weekly cumulative IDF curves to identify distinct engagement trajectories over the term.

5 Methods

5.1 Primary research questions

Two substantive questions guide the analysis reported in Sections 4.2 and 4.3. **RQ1 (component importance):** Which of Frequency, Immediacy and Diversity contributes most strongly to explaining variation in final grades? Because the three indicators are highly correlated (mean pairwise $|r| \approx 0.75\text{--}0.83$; $\text{VIF} \approx 2.6\text{--}4.4$), ordinary partial regression coefficients are unstable. The LMG relative-importance decomposition (Lindeman et al., 1980) partitions model R^2 into fair shares that are stable under collinearity, computed per cohort at weeks 6 (mid-term) and 11 (end-of-term). Bivariate rank correlations from meta-analysis provide a contrast: they show whether each indicator associates with grade in isolation, without removing shared variance.

RQ2 (heterogeneity across the grade distribution): Does engagement affect all students equally? Quantile regression (Koenker and Bassett, 1978) estimates the relationship between standardised cumulative engagement and final grade at selected quantiles $\tau \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$, interpreted as the grade-point change per one standard deviation increase in engagement among students at the τ -quantile of the grade distribution. The contrast between $\tau = 0.1$ (lower tail) and $\tau = 0.5$ (median) is summarised descriptively and with student-level bootstrap confidence intervals (500 replicates per cohort). Univariate quantile models (one indicator at a time) identify which component drives lower-tail associations. Analyses use weeks 6 and 11.

5.2 Association between engagement and final grade

Within each cohort, Spearman rank correlations were computed between cumulative engagement and final grade at each week and at the end of term. To synthesise evidence across cohorts, cohort-

specific correlations were pooled using a random-effects meta-analysis, reporting the pooled correlation, confidence interval and I^2 heterogeneity statistic. Complementing rank-based evidence, a mixed-effects model was fitted with final grade as the outcome and standardised cumulative IDF as the predictor, allowing the slope to vary by cohort. This expresses the association on the grade scale in points per one standard deviation of engagement.

5.3 Timing and early-course usefulness

Weekly Spearman correlations between cumulative engagement and final grade were tracked across the term to assess how early the association emerges. For each cohort, a bootstrap procedure identified the earliest week at which the correlation was statistically indistinguishable from its end-of-term value, providing a practical sense of when the signal stabilises.

5.4 Lower-tail and early-warning analysis

Students were grouped into engagement quintiles at selected weeks, and the distribution of final grades within each quintile was examined. Quantile regression estimates the relationship between engagement and final grade at different points in the grade distribution—for example, among the lowest 10% of performers rather than only at the median—so that one can ask whether the metric matters more for students already at academic risk. Particular attention was given to the lower tail ($\tau = 0.1$) relative to the median ($\tau = 0.5$). For early-warning relevance, the bottom engagement quintile was treated as a risk flag and evaluated using the area under the ROC curve (AUC), recall and precision for identifying students scoring below 50% at the end of term.

5.5 Diagnostic decomposition of engagement

Frequency, Immediacy and Diversity are strongly correlated, so ordinary regression coefficients do not reveal which dimension carries unique information. Variance inflation factors were examined to confirm collinearity, and the LMG relative-importance decomposition was used to partition explained variance among the three indicators in a way that is stable under collinearity: it averages each predictor’s contribution to R^2 over all possible orderings in which the indicators enter the model.

5.6 Assessment-component validity

Where separate assessment marks were available, engagement was correlated with each component and Steiger’s test for dependent correlations was used to compare whether engagement aligns more strongly with individually assessed work (exams, quizzes) than with group- or peer-marked components.

5.7 Temporal regularity and trajectory analysis

Timing and regularity features were added to models already containing F, I and D to test whether *how* students engage over time adds information beyond *how much* and *how broadly* they engage. Separately, latent-class mixed models were applied to weekly cumulative IDF profiles to identify groups of students following distinct engagement trajectories and to compare their final grades. To assess operational usefulness, end-of-term class labels were compared with early assignments made by matching each student’s partial weekly profile (weeks 1– w) to the corresponding segment of the end-of-term class centroids.

5.8 Robustness checks

The main conclusions were examined under degree-programme adjustment, false-discovery-rate correction using the Benjamini–Hochberg procedure across weekly association tests, alternative bounded outcome specifications, and variation in the at-risk engagement threshold. These checks were kept brief and interpreted in the context of the primary findings rather than treated as a separate technical audit.

6 Results

The results open with overall engagement–grade association (Section 4.1), then address the two primary research questions in detail: component importance (Section 4.2) and heterogeneity across the grade distribution (Section 4.3). Supporting diagnostics, practical limits and robustness follow.

6.1 Overall engagement–grade association

Across all five cohorts, cumulative engagement is positively and moderately associated with final grade. The random-effects pooled rank correlation for cumulative IDF at week 11 is $r = 0.32$ (95% CI [0.25, 0.38]), with low-to-moderate heterogeneity across cohorts ($I^2 = 39\%$). Frequency, Immediacy and Diversity show similar pooled associations when analysed separately (Table 2). On the grade scale, the mixed-effects estimate corresponds to approximately 3.8 grade points per one standard deviation increase in engagement. Figure 1 shows that every cohort lies above zero, but the association is far from perfect prediction—engagement is a useful but imperfect proxy for academic performance.

Table 2: Pooled rank correlations and mixed-effects slopes at week 11.

Indicator	Pooled r [95% CI]	I^2	Slope (SE)	Moderator p
Frequency	0.28 [0.22, 0.34]	27%	3.36 (0.53)	0.92
Immediacy	0.30 [0.24, 0.37]	37%	3.63 (0.58)	0.88
Diversity	0.31 [0.24, 0.37]	33%	3.62 (0.53)	0.87
IDF	0.32 [0.25, 0.38]	39%	3.79 (0.58)	0.89

Source: author’s analysis of UCL Moodle log data.

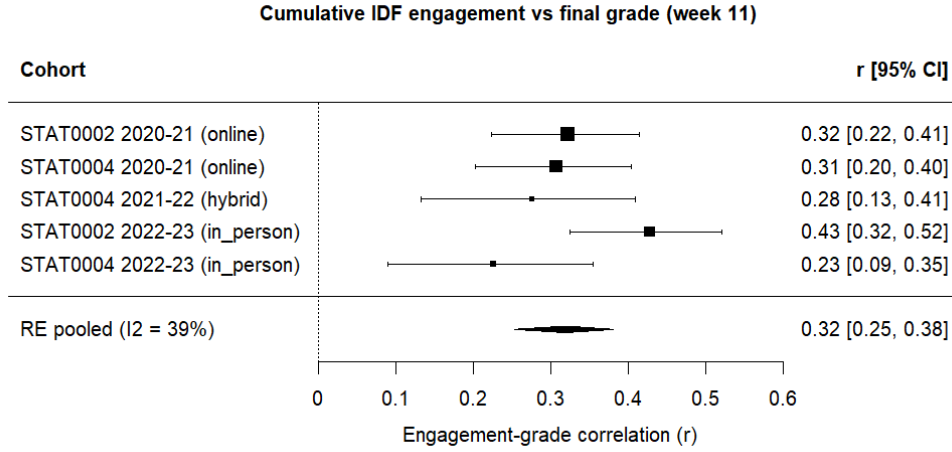


Fig. 1: Cohort-specific and pooled week-11 correlations between cumulative IDF engagement and final grade. All cohorts show positive associations; the random-effects estimate summarises the overall moderate engagement–grade relationship.

STAT0002 deliveries show stronger cohort-level associations than STAT0004 ($r = 0.28$ – 0.41 versus $r = 0.19$ – 0.30), consistent with the module-level patterns noted in Section 3. The pooled estimate nonetheless remains positive and moderate because every cohort contributes in the same direction.

When students are ranked by cumulative engagement and grouped into quintiles, final grades tend to rise across successive quintiles, particularly in STAT0002, where quintile separation is clearest (Figure 2). At weeks 3 and 6, the lowest-engagement quintile is the group whose grade distribution most often approaches or crosses the 50% low-performance threshold, while the highest-engagement quintile consistently shows the strongest outcomes. Separation between adjacent middle quintiles is weaker, which is consistent with a monotone but noisy relationship rather than sharp boundaries between moderate engagement levels. Together, the pooled correlation, positive cohort-level associations and quintile patterns support the conclusion that the metric *travels* to these new module-year contexts with a moderate, robust association.

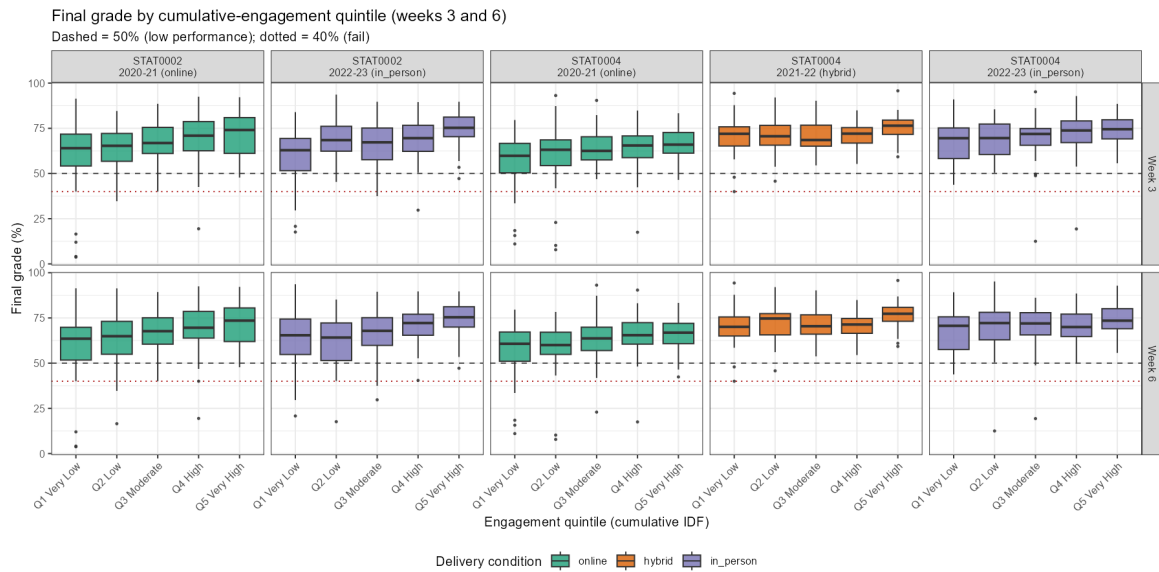


Fig. 2: Final-grade distributions by engagement quintile at weeks 3 and 6.

6.2 RQ1: Which engagement components matter most?

Treating engagement as a single composite score confirms that it relates to grades, but obscures *which* learning behaviours carry unique predictive information. Frequency (how often students interact), Diversity (the variety of resource types accessed) and Immediacy (how promptly students engage with newly released content) are strongly correlated within cohorts: mean pairwise $|r| \approx 0.75\text{--}0.83$, with variance inflation factors between 2.6 and 4.4 at weeks 6 and 11. Under such collinearity, ordinary multiple-regression coefficients are unstable and can even change sign across cohorts; a method that partitions explained variance fairly is required.

6.2.1 Method and bivariate contrast

The LMG decomposition averages each predictor’s contribution to R^2 over all orderings in which Frequency, Immediacy and Diversity enter the model, yielding a stable relative-importance share per indicator (Lindeman et al., 1980). As a contrast, bivariate pooled rank correlations at week 11 show similar associations for all three indicators when analysed separately: $r = 0.28$ (Frequency), $r = 0.30$ (Immediacy) and $r = 0.31$ (Diversity; Table 2). Taken alone, this would suggest that all components matter equally. The LMG results show that this is misleading once shared variance is removed.

6.2.2 Results at week 11

At end-of-term (week 11), Diversity accounts for the largest mean share of explained variance (37%), Immediacy is intermediate (36%), and Frequency the smallest (27%). Frequency has the smallest LMG share in every cohort (Table 3, Figure 3). Cohort-level patterns vary: Immediacy dominates in STAT0002 online (45%), whereas Diversity dominates in STAT0004 online and in-person (45% and 42% respectively).

Table 3: LMG relative-importance shares (%) at week 11.

Cohort	Frequency	Immediacy	Diversity
STAT0002 online	28	45	27
STAT0002 in-person	28	36	36
STAT0004 online	23	32	45
STAT0004 hybrid	31	32	38
STAT0004 in-person	25	33	42
Mean	27	36	37

Source: author’s analysis of UCL Moodle log data.

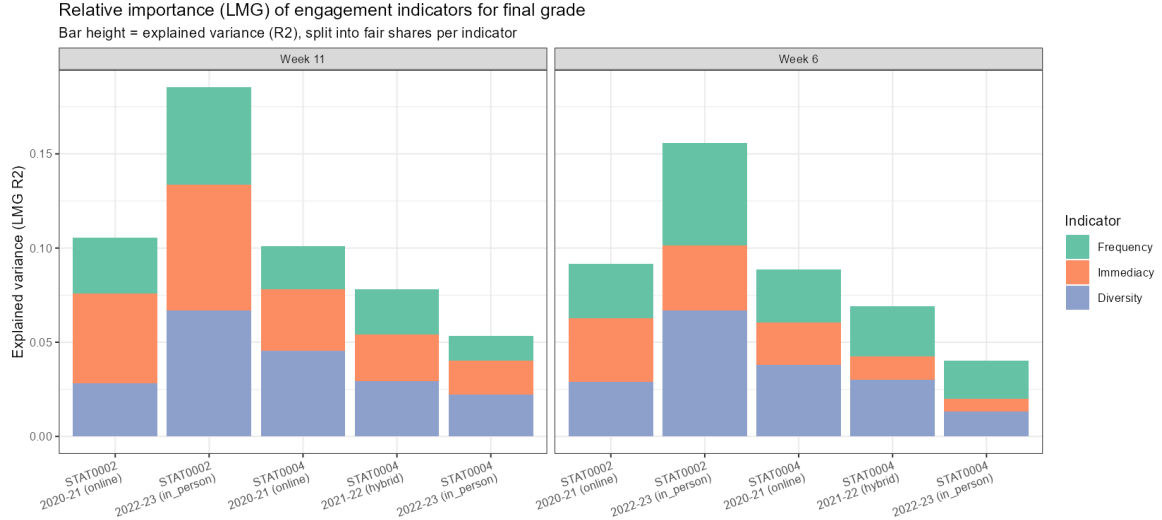


Fig. 3: LMG relative-importance shares for Frequency, Immediacy and Diversity by cohort at weeks 6 (left) and 11 (right). Diversity and Immediacy typically account for more unique explained variance than Frequency.

6.2.3 Stability at mid-term (week 6)

The same decomposition at week 6—the course midpoint and a common decision point for early support—shows a broadly similar ranking: Diversity remains the top contributor on average (39%), with Immediacy lowest (24%) and Frequency intermediate (37%). One cohort (STAT0004 in-person) assigns a larger week 6 share to Frequency (50%), but Frequency never ranks first at week 11 in any cohort. The pooled pattern supports interpreting breadth of resource use as the most consistent unique signal, with promptness also carrying substantial information, especially in first-year STAT0002.

Table 4: LMG relative-importance shares (%) at week 6.

Cohort	Frequency	Immediacy	Diversity
STAT0002 online	31	37	32
STAT0002 in-person	35	22	43
STAT0004 online	32	25	43
STAT0004 hybrid	38	18	43
STAT0004 in-person	50	17	33
Mean	37	24	39

Source: author's analysis of UCL Moodle log data.

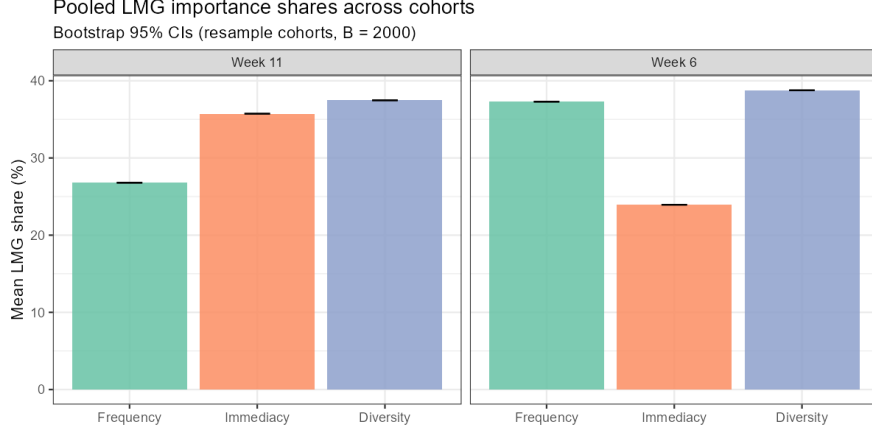


Fig. 4: Pooled mean LMG importance shares across cohorts at weeks 6 and 11.

6.2.4 Interpretation

In the engagement literature, behavioural engagement is not reducible to time-on-task: it reflects participation breadth, effort and timely involvement with academic work (Fredricks et al., 2004). Frequency counts log events and is sensitive to superficial or repeated access without sustained study. Diversity captures whether students explore the full range of chapter resources—curriculum coverage rather than revisiting a single file. Immediacy reflects alignment with the release schedule, consistent with staying current rather than falling behind. The LMG results indicate that outcome relevance lies in indicators of structured study behaviour, not raw click volume. **Takeaway:** engagement matters, but breadth and promptness carry more unique signal than session frequency; bivariate correlations alone would not reveal this distinction.

6.3 RQ2: Does engagement affect all students equally?

A single average association—whether from OLS or Spearman correlation—can hide whether engagement is more consequential for students already at academic risk. Quantile regression addresses this by estimating separate slopes at different points in the grade distribution.

6.3.1 Method

Models take the form $\text{grade} = \beta_0 + \beta_1(\tau) z_{\text{IDF}} + \varepsilon$, where z_{IDF} is within-cohort standardised cumulative engagement and $\beta_1(\tau)$ is the slope at quantile τ of the grade distribution. Positive $\beta_1(\tau)$ means that higher engagement is associated with higher grades among students at that performance level. If engagement is a **support tool for at-risk learners**, slopes should be largest at low τ ; if it is a **universal predictor**, slopes should be similar across τ .

6.3.2 Results: composite engagement (IDF)

At week 11, the IDF slope is steeper at $\tau = 0.1$ than at $\tau = 0.5$ in **every cohort** (Table 5). On average, the lower-tail slope exceeds the median slope by 2.3 grade points per standard deviation of engagement. Slopes generally decrease from $\tau = 0.1$ through $\tau = 0.5$ to $\tau = 0.9$, indicating that engagement variation is most consequential among lower-performing students. For example, in STAT0002 in-person the slope is 7.9 points at $\tau = 0.1$ versus 5.0 at the median and 3.0 at $\tau = 0.9$. STAT0004 hybrid shows the smallest tail excess (0.6 points), suggesting cohort-specific heterogeneity in how sharply the lower tail diverges.

Student-level bootstrap confidence intervals for $\beta_1(0.1) - \beta_1(0.5)$ confirm the pattern in two of five cohorts at week 11 (STAT0002 and STAT0004 in-person); in the remaining cohorts the point estimates are positive but intervals are wide (Table 6). At week 6, three cohorts show positive descriptive contrasts and one (STAT0004 in-person) has a bootstrap interval excluding zero.

Table 5: Quantile-regression slopes for cumulative IDF at week 11 (grade points per +1 SD).

Cohort	$\tau = 0.1$	$\tau = 0.5$	$\tau = 0.9$	$\tau_{0.1} - \tau_{0.5}$
STAT0002 online	5.32	3.22	2.58	2.11
STAT0002 in-person	7.88	5.04	2.96	2.84
STAT0004 online	6.06	3.15	2.17	2.91
STAT0004 hybrid	3.26	2.66	1.34	0.59
STAT0004 in-person	3.97	1.10	1.28	2.88

Source: author's analysis of UCL Moodle log data.

Table 6: Bootstrap 95% CI for $\tau_{0.1} - \tau_{0.5}$ contrast (week 11).

Cohort	Contrast	95% CI	CI > 0?
STAT0002 online	2.11	[-1.60, 4.79]	No
STAT0002 in-person	2.84	[0.16, 5.54]	Yes
STAT0004 online	2.91	[-0.85, 4.77]	No
STAT0004 hybrid	0.59	[-2.62, 4.49]	No
STAT0004 in-person	2.88	[0.26, 5.77]	Yes

Source: author's analysis of UCL Moodle log data.

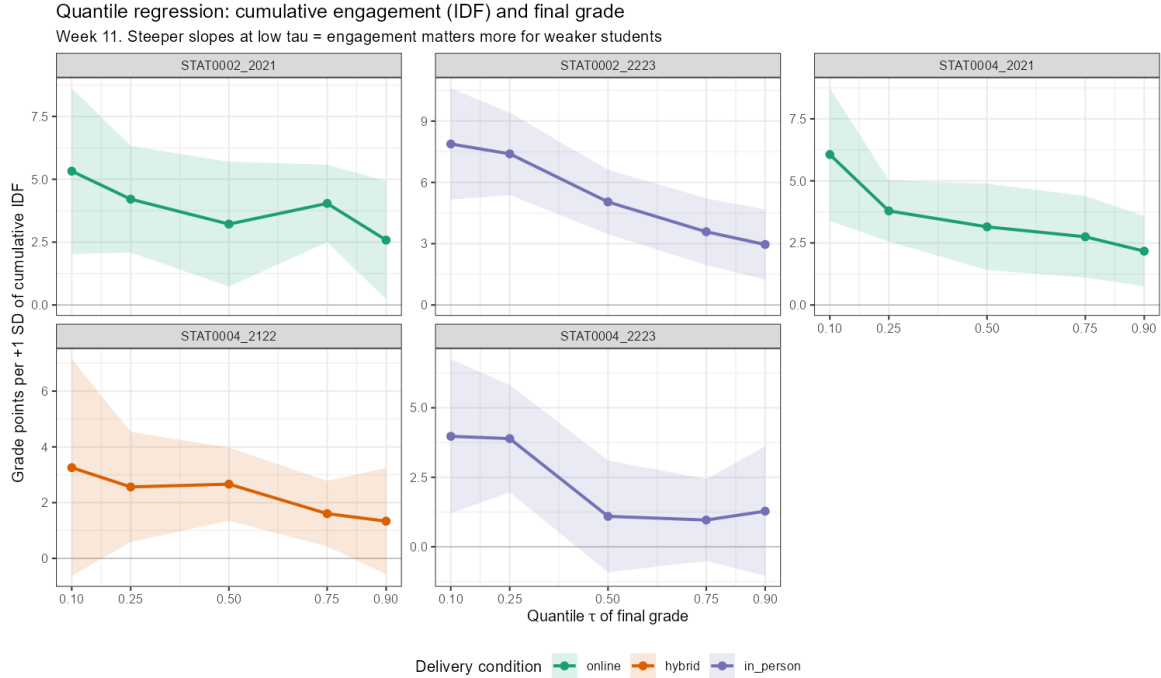


Fig. 5: Quantile-regression slopes of final grade on cumulative IDF at week 11, with 95% confidence intervals, by cohort.

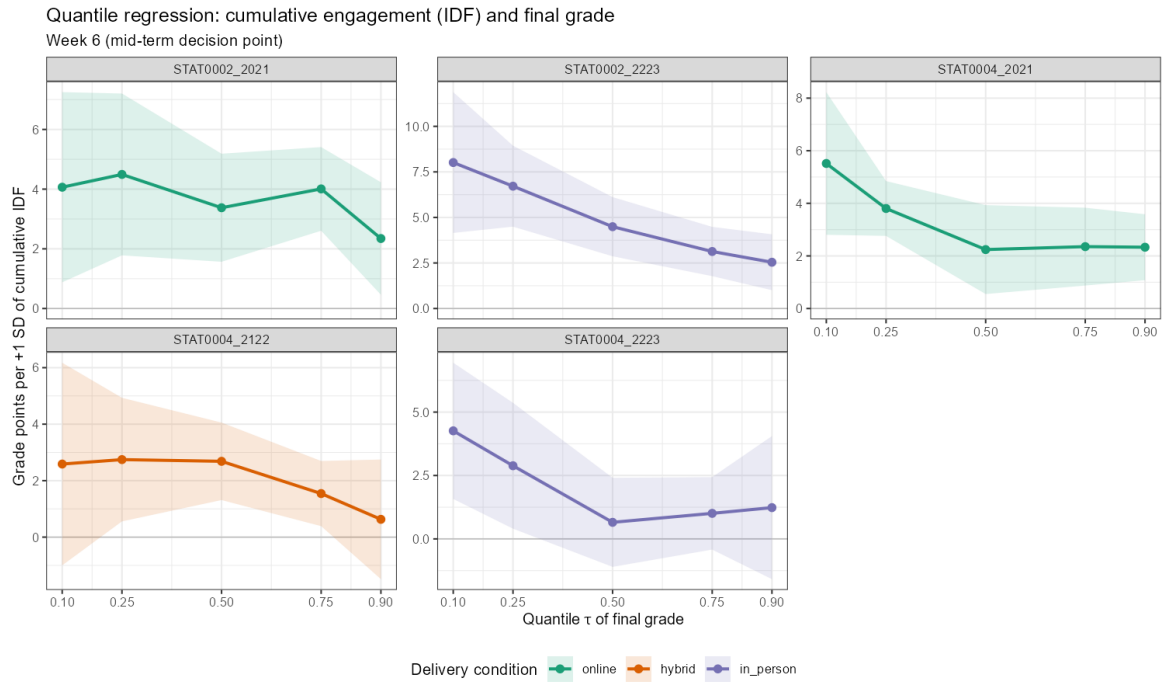


Fig. 6: Quantile-regression slopes at week 6 (mid-term). The lower-tail pattern is visible in most cohorts by the course midpoint.

6.3.3 Which component drives lower-tail effects?

Univariate quantile regression (one indicator at a time) shows that **Immediacy** has the largest mean slope at $\tau = 0.1$ across cohorts at week 11 (4.9 grade points per SD), followed by Diversity and Frequency (Figure 7). This complements the LMG finding that Diversity carries the most unique variance in multivariate models: promptness appears especially salient for distinguishing outcomes among lower-performing students when indicators are considered separately.

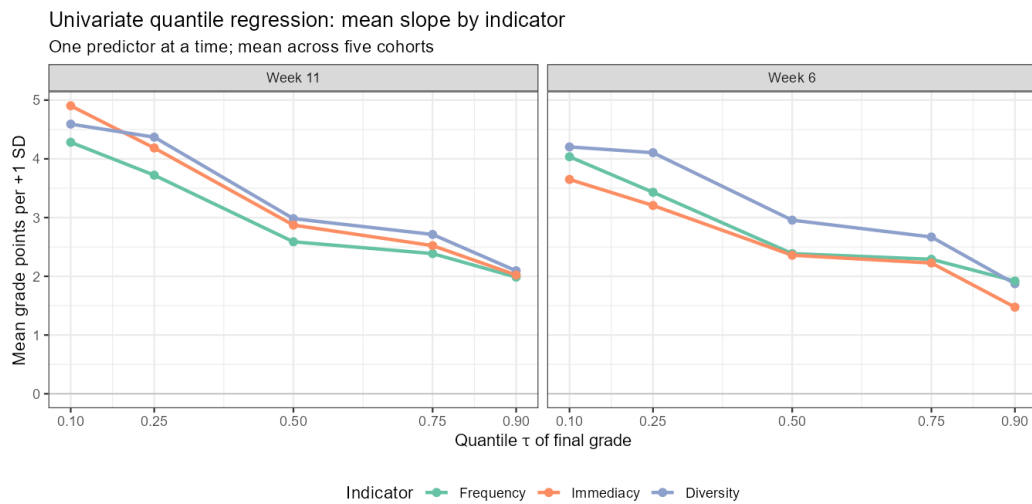


Fig. 7: Mean univariate quantile-regression slopes by indicator (weeks 6 and 11; mean across five cohorts). Larger coefficients at low τ indicate stronger associations among lower-performing students.

6.3.4 Interpretation

The quantile results support viewing engagement as a **support tool for at-risk learners** rather than a predictor with uniform effect across the ability distribution. Interventions that raise engagement among disengaged, lower-performing students are associated with the largest grade differences; among already high-performing students, additional engagement variation is less consequential. **Takeaway:** engagement variation is most valuable for identifying and supporting academically at-risk students, not for ranking the entire cohort with equal precision.

6.4 Supporting diagnostics: trajectories, temporal patterns and assessment validity

Bootstrap stabilisation analysis asks whether additional weeks materially change the engagement–grade correlation, not whether the correlation is already strong (Table 7, Figure 8). In four cohorts, the weekly correlation is statistically close to its end-of-term value from week 1 or 2, but this sometimes reflects a relatively flat and modest signal rather than impressive early predictive power. For example, STAT0004 in-person stabilises early yet has a weak end-of-term association ($r = 0.19$); STAT0004 hybrid similarly stabilises at week 1 with $r = 0.23$. By contrast, STAT0002 in-person shows a strengthening association that does not stabilise until around week 9, reaching the strongest cohort-level correlation ($r \approx 0.41$), suggesting that later engagement adds meaningful information in that delivery. Stabilisation timing and association strength should therefore be read separately: early stabilisation does not guarantee a strong early-warning signal.

Table 7: Stabilisation week and end-of-term IDF–grade correlation.

Cohort	Stabilisation week	End-of-term r
STAT0002 online	2	0.28
STAT0002 in-person	9	0.41
STAT0004 online	1	0.30
STAT0004 hybrid	1	0.23
STAT0004 in-person	1	0.19

Source: author’s analysis of UCL Moodle log data.

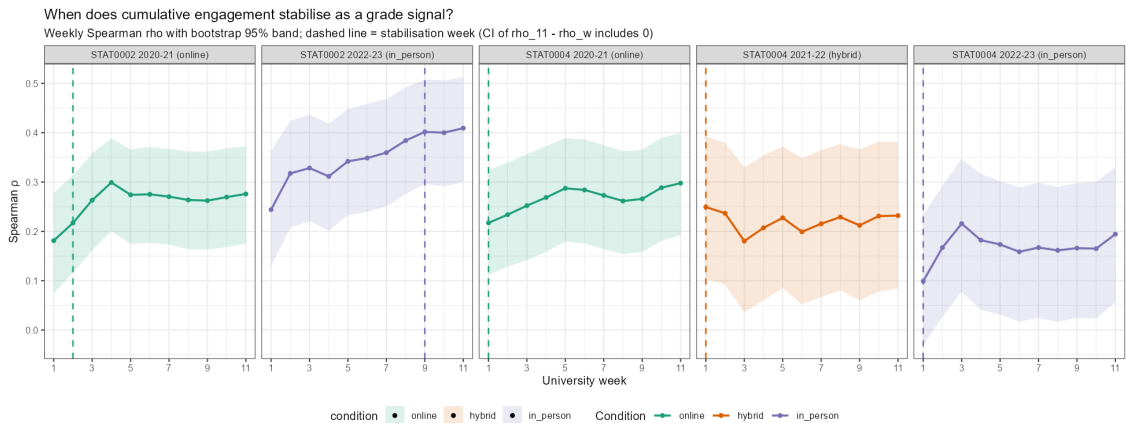


Fig. 8: Weekly Spearman correlation between cumulative engagement and final grade by cohort, with bootstrap stabilisation week marked. Early stabilisation can reflect a flat, modest signal as well as a strong one; compare stabilisation week with end-of-term r in Table 7.

Latent-class analysis of weekly cumulative IDF profiles identifies interpretable trajectory types—low-steady, high-steady and mid-variable (Figure 9)—and final grades differ markedly across

classes (Kruskal–Wallis $p = 2.6 \times 10^{-17}$). The high-steady class averages 72.9, compared with 63.2 for the low-steady class: a gap of about 9.7 grade points. The low-steady class also contains the highest share of low performers (15.7%). Sustained engagement *patterns* over the term therefore separate outcomes more sharply than a single cumulative score at one time point, and identify a group whose profile is persistently low rather than temporarily quiet.

Table 8: Trajectory classes and grade outcomes ($N = 1,290$).

Class	N	Mean grade	% < 50
Low-steady	439	63.2	15.7
High-steady	165	72.9	2.4
Mid-variable	686	68.3	5.5

Source: author’s analysis of UCL Moodle log data.

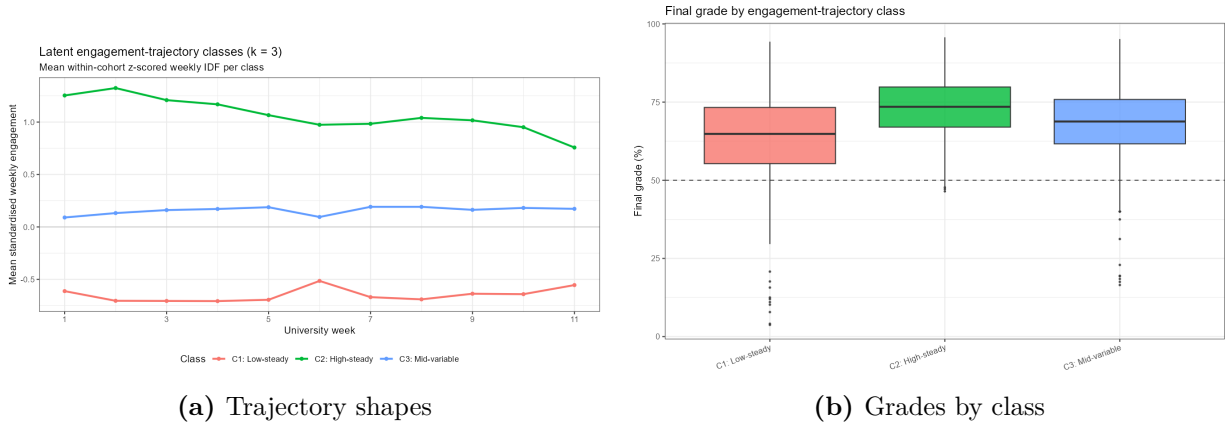


Fig. 9: Latent engagement trajectories (left) and final-grade distributions by trajectory class (right). Three trajectory types separate outcomes by roughly 9.7 grade points between the highest- and lowest-engagement profiles.

A natural operational question is how early a student can be assigned to a trajectory class with reasonable confidence. End-of-term class labels were therefore compared with early assignments made by matching each student’s partial weekly profile (weeks 1– w) to the corresponding segment of the end-of-term class centroids (Table 9, Figure 10). Overall agreement rises from 71% at week 3 to 90% by week 9. For the low-steady class specifically, recall reaches 81% by week 4 and 88% by week 6; precision that an early low-steady flag corresponds to the true end-of-term low-steady class is 77% at week 4 and 81% at week 6. These levels are materially higher than the bottom-quintile flag’s precision for identifying students who ultimately score below 50% (about 15–26%). However, among students flagged as low-steady early, only about 15% still score below 50%—similar to the quintile rule. Trajectory class is therefore a more reliable early label for *sustained low engagement* than the quintile is for *eventual failure*: operationally useful for targeted, pattern-based outreach from mid-term onwards, but not a sharp failure predictor.

Table 9: Early recovery of end-of-term trajectory class from partial weekly profiles.

Week	Agreement (%)	Low-steady recall (%)	Low-steady precision (%)	% flagged < 50
3	70.9	78.6	69.0	14.8
4	76.9	81.3	76.8	14.6
6	81.6	88.2	80.8	14.8

Source: author’s analysis of UCL Moodle log data.

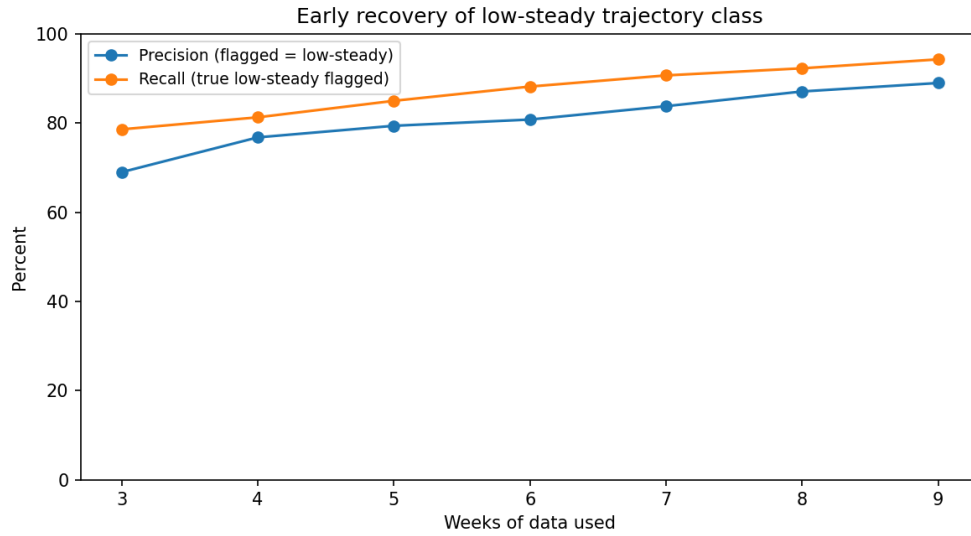


Fig. 10: Recall and precision for early assignment to the low-steady trajectory class as more weekly data accumulate. Agreement improves steadily from week 3 onwards.

Engagement correlates more strongly with individually assessed components than with group- or peer-marked work where both are available (Figure 11). In STAT0002 in-person, the correlation with the exam grade ($r = 0.40$) is significantly larger than with peer-marking ($r = 0.20$; Steiger $z = 2.98$, $p = 0.003$). Other component contrasts point in the same direction but are not statistically significant. This supports the interpretation that Moodle engagement reflects individual study behaviour more faithfully than outcomes shaped by group dynamics or peer assessment.

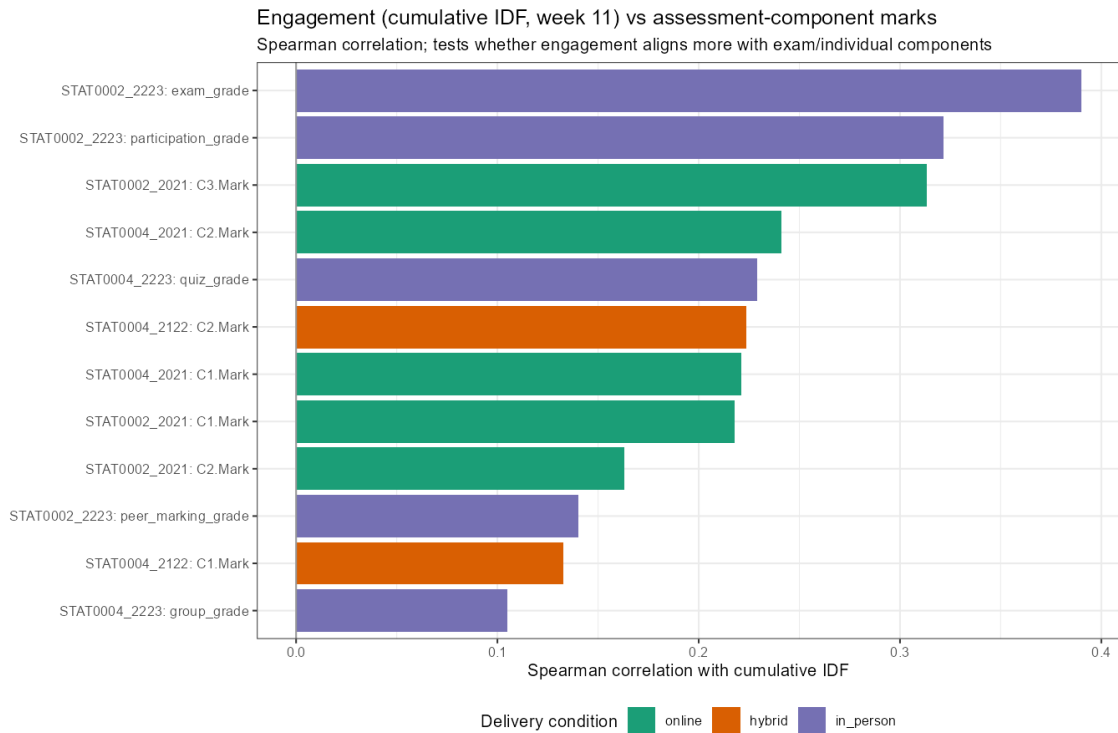


Fig. 11: Engagement-assessment component correlations by cohort.

Timing and regularity features add incremental explanatory power beyond F, I and D in the

STAT0002 cohorts, where engagement showed stronger alignment with individual assessment components (Table 10, Figure 12). At week 6, the gain in R^2 is about 0.05 in both STAT0002 deliveries ($p = 0.003$ and $p = 0.006$), but not statistically significant in the STAT0004 deliveries. How regularly students work carries additional diagnostic information in STAT0002, beyond what the core composite already captures.

Table 10: Incremental R^2 from temporal features at week 6.

Cohort	R^2 (F/I/D)	R^2 (full)	ΔR^2	p
STAT0002 online	0.091	0.143	0.052	0.003
STAT0002 in-person	0.156	0.213	0.057	0.006
STAT0004 online	0.089	0.094	0.005	0.939
STAT0004 hybrid	0.069	0.102	0.033	0.424
STAT0004 in-person	0.040	0.094	0.053	0.094

Source: author’s analysis of UCL Moodle log data.

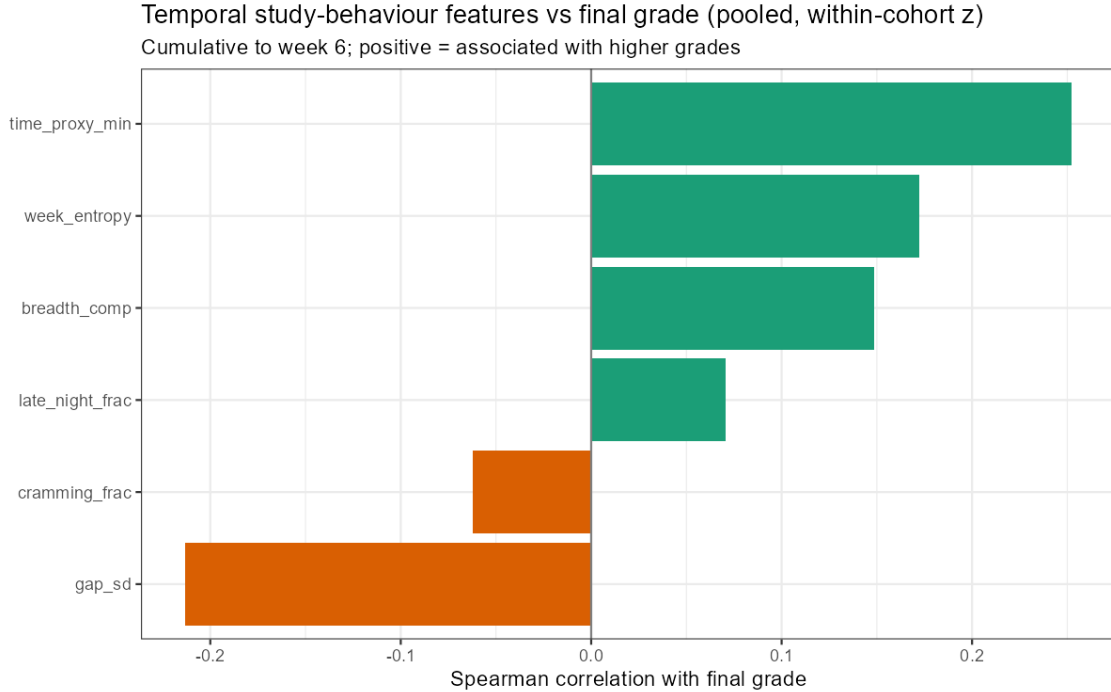


Fig. 12: Temporal and regularity features correlated with final grade.

The diagnostic extensions are not uniform across modules. Temporal regularity adds significant incremental R^2 only in STAT0002, where engagement also aligns more strongly with individually assessed components. In STAT0004, where associations are weaker and group or coursework components carry more weight, the composite and its extensions explain less outcome variance—consistent with greater off-platform study, second-year autonomy, or looser chapter–assessment alignment weakening the Moodle signal rather than invalidating the metric outright.

6.5 Practical limits: early-warning precision

Although quantile regression shows greater outcome relevance in the lower grade tail (Section 4.3), operational flagging still faces modest precision. At the course midpoint, the bottom engagement quintile in STAT0002 identifies roughly 38–47% of students who ultimately score below 50% (AUC around 0.71–0.76), but precision is only about 15–26%: roughly three-quarters to four-

fifths of flagged students may still pass. Figure 13 illustrates this recall–precision trade-off. The metric is therefore useful for identifying students with low *observed* Moodle engagement, not for identifying students who will fail—appropriate responses are broad, low-cost and supportive rather than intensive or punitive.

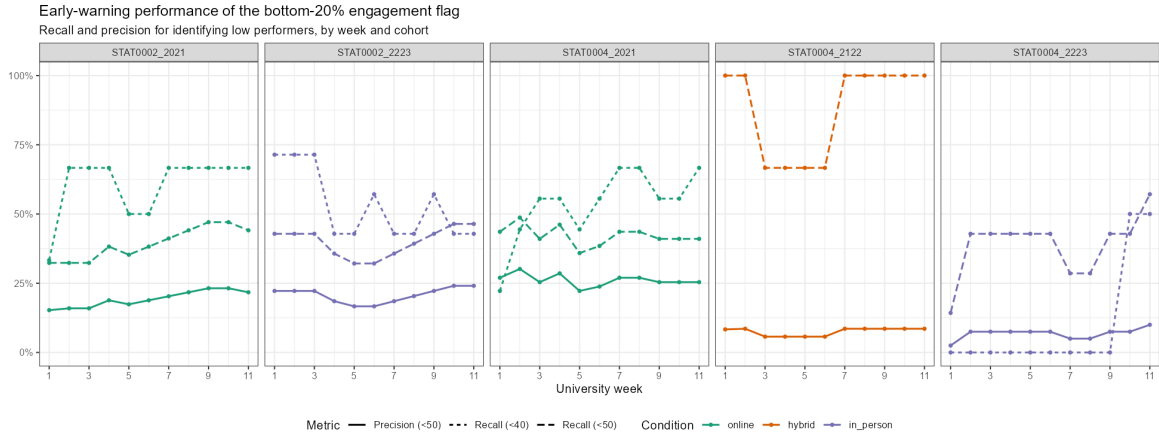


Fig. 13: Recall and precision for the bottom engagement quintile across weeks. Modest precision implies that most flagged students may still achieve passing grades.

6.6 Boundary condition: delivery period and cohort confounding

Because delivery condition is inseparable from cohort and year in this dataset, comparisons across online, hybrid and in-person periods are best treated as a descriptive boundary check rather than a test of delivery effects. Meta-analytic moderator tests do not suggest that delivery label explains between-cohort variation (all $p > 0.87$; Table 2), and weekly correlations appear broadly overlapping across periods (Figure 14). This should not be interpreted as evidence that delivery mode is irrelevant. Module design, assessment structure, cohort composition and Moodle resource organisation may be at least as important as the delivery label itself.

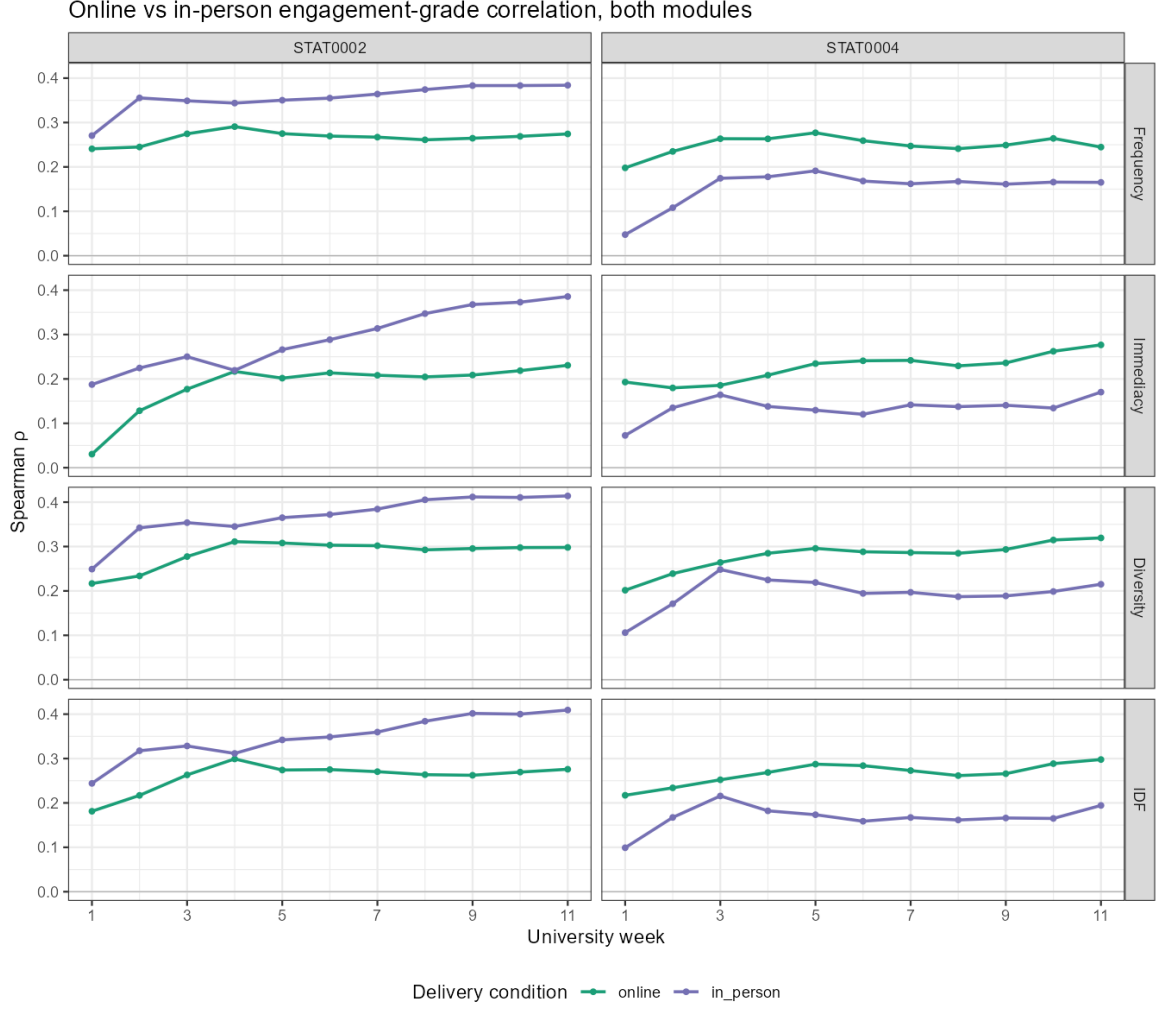


Fig. 14: Weekly engagement–grade correlation by module and delivery period. Comparisons are descriptive; condition is confounded with academic year.

6.7 Robustness checks

Each check addressed a specific threat to the headline conclusions. Programme adjustment tests whether the pooled engagement slope reflects degree-programme composition rather than study behaviour itself; the slope changes by at most 14%, so programme mix is not driving the result. False-discovery-rate correction using the Benjamini–Hochberg procedure guards against spurious weekly associations from repeated testing across weeks and indicators; 205 of 220 tests remain significant. Alternative bounded-outcome specifications address skew and ceiling effects in grade distributions and agree in sign with the main models in all five cohorts. Finally, varying the at-risk engagement threshold examines whether early-warning conclusions depend on an arbitrary quintile cut-off; recall and precision shift in the expected direction without changing the overall interpretation.

7 Discussion

The results support three linked conclusions, with the first two corresponding to the primary research questions developed in Sections 4.2 and 4.3.

RQ1 (component importance). Once collinearity among Frequency, Immediacy and Di-

iversity is accounted for, Diversity and Immediacy carry more unique information than Frequency (mean LMG shares at week 11: 37%, 36% and 27%). The ranking is broadly stable at week 6, when practitioners might first act on engagement data. Bivariate correlations would misleadingly suggest all three matter equally ($r \approx 0.28\text{--}0.31$); only decomposition reveals that raw click volume contributes least. This aligns with Fredricks et al. (2004)’s distinction between behavioural engagement as structured participation and mere time-on-task.

RQ2 (heterogeneity across the grade distribution). Quantile regression shows that engagement–grade associations are strongest among lower-performing students: in every cohort the slope at $\tau = 0.1$ exceeds that at the median, with a mean excess of 2.3 grade points per standard deviation of engagement. Engagement is therefore best understood as a *support tool for at-risk learners* rather than a uniform predictor across ability levels. Univariate quantile models suggest Immediacy is especially salient in the lower tail when indicators are considered separately, complementing the multivariate LMG finding that Diversity carries the most unique variance overall.

External validation. The chapter-aligned metric remains a moderate, portable engagement signal in new STAT0002 and STAT0004 cohorts: pooled $r = 0.32$, a mixed-effects slope of roughly 3.8 grade points per standard deviation, positive associations in every cohort, and clear quintile separation in grade outcomes. This extends the primary findings of Johnston et al. (2025) through pooled inference across module-year contexts not used in the original validation study.

Supporting diagnostics. Sustained engagement trajectories separate final grades by roughly 9.7 points between low-steady and high-steady profiles; partial weekly profiles recover the low-steady class with about 81% recall by week 4. Temporal regularity adds explanatory power in STAT0002 but not STAT0004, and engagement aligns more closely with individually assessed work than with peer- or group-marked components.

Practical limits. The association is concentrated in the lower grade tail, but bottom-quintile precision is modest (about 15–26%). The metric identifies students with low *observed* Moodle engagement, not students who will fail. Delivery-period comparisons provide no evidence of a large moderation effect by delivery label, but because condition is perfectly confounded with academic year and cohort, they cannot establish that delivery mode is irrelevant.

8 Practical implications

The bottom-engagement flag should be interpreted as identifying students with low *observed* Moodle engagement, not as identifying students who will fail. With precision of about 15–26% at the course midpoint in STAT0002, roughly 74–85% of flagged students may still achieve passing grades. That does not make the metric useless, but it changes how it should be used.

Appropriate uses include broad, low-cost outreach: study reminders, non-accusatory check-ins (“we noticed you may not have accessed much material”), invitations to office hours, prioritising students for human review, and dashboard-level monitoring of chapters and resources. Engagement should serve as a *first-pass* signal, combined with coursework marks, attendance where available, tutor knowledge and student self-report before any substantive intervention.

Inappropriate uses include labelling a student as failing, punitive emails, formal escalation on engagement alone, high-stakes ranking, replacing tutor judgement, or assuming low Moodle engagement means low effort. Low engagement may reflect access barriers, competing commitments or study strategies that do not register fully in logs.

When engagement is low, the more useful question is *why* rather than *whether*: is the student behind the release schedule (Immediacy), touching only a narrow set of resources (Diversity), showing low temporal regularity, or following a persistently low trajectory? From week 4 onwards, partial weekly profiles can assign students to a low-steady trajectory class with roughly 80%

recall and 77% precision—a stronger basis for pattern-based outreach (“your engagement has been consistently low across the term so far”) than the bottom quintile alone. Those distinctions guide how support is offered. Any operational use in a new module or delivery context should be preceded by local validation of thresholds and association strength, especially where assessment design or resource labelling differs from the cohorts studied here.

9 Limitations

Several limitations should be kept in view. Delivery condition is confounded with academic year and cohort, so no causal inference about online, hybrid or in-person delivery is possible. The analysis covers only two statistics modules and five deliveries, limiting generalisability. Moodle logs capture online behavioural engagement only; offline study is invisible. Final grades are imperfect outcome measures, especially where group, peer or non-invigilated assessment contributes. Low-performer counts are small in some cohorts, making early-warning metrics unstable. The metric depends on consistent chapter labelling in Moodle. All findings are associational, not causal. Frequency, Immediacy and Diversity are highly collinear, so indicator-level interpretation requires appropriate methods. Finally, external validation would be needed before operational deployment in new contexts without recalibration.

10 Conclusion

This report validates an existing real-time Moodle engagement metric on new STAT0002 and STAT0004 cohorts spanning online, hybrid and in-person delivery periods. Cumulative engagement is moderately associated with final grades ($r = 0.32$ pooled). Two findings receive particular emphasis. **First**, decomposing engagement into Frequency, Immediacy and Diversity shows that breadth and promptness of resource use outweigh raw session frequency once indicator overlap is removed—a distinction that bivariate correlations alone would not reveal. **Second**, quantile regression shows that engagement matters more for lower-performing students than for the average student, supporting targeted support for at-risk learners rather than uniform prediction across the grade distribution. Trajectory classes, temporal regularity and assessment-component alignment provide additional diagnostic context. Delivery comparisons are descriptive only. The metric should be used for low-cost supportive monitoring with local validation, not high-stakes prediction.

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